

## QUIZ 02 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

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### ⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Wednesday. Rather, they are designed to give you a sense of the style of multiple choice questions I typically ask. As a reminder, the quiz on Wednesday will be 15 questions.

### Use the following setup for Questions 1 through 4:

Jacob wants to determine the proportion of apartments in California that are one-bedroom apartments. To that end, he goes to *Zillow* and sets his search radius to San Luis Obispo. He then assigns a unique number to each of the apartments listed, uses a random number generator to generate 10 numbers, and looks up the apartments corresponding to these 10 numbers. His 10 sampled apartments contained 4 one-bedroom apartments.

1. Describe an observational unit in the context of this dataset.

- (A) One person
- (B) One apartment**
- (C) One one-bedroom apartment
- (D) Jacob
- (E) San Luis Obispo

**Solution:** Jacob is collecting information (specifically the “one-bedroom status”) from individual apartments - as such, one observational unit is **one apartment**.

2. What is the sampling frame in the context of this study?

- (A) The set of all California apartments
- (B) The set of all California apartments listed on *Zillow*
- (C) The set of all San Luis Obispo apartments
- (D) The set of all San Luis Obispo apartments listed on *Zillow***

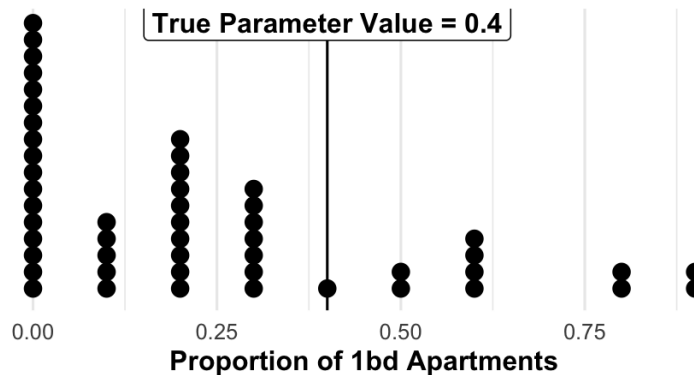
**Solution:** Jacob will only ever sample San Luis Obispo apartments listed on *Zillow*, meaning his sampling frame is the **set of all SLO apartments listed on *Zillow***.

3. The value of 4/10 is important to this problem. Is it a parameter value or a statistic value?

- (A) Parameter
- (B) Statistic**

**Solution:** The value of 4/10 is the sample proportion of apartments that were one-bedroom.

4. Suppose Jacob has carried out his sampling procedure 50 times, each time taking a sample of 10 apartments and recording the proportion that are one-bedroom. Shown below is a dotplot of his statistic values, along with the true value of the parameter (which he did not know in advance):



**Figure 1:** Dotplot of Jacob's 50 sampled proportions.

Based on this graph, do you think Jacob's sampling method is unbiased, biased with a tendency to underestimate, or biased with a tendency to overestimate?

- (A) Unbiased
- (B) Biased, with a tendency to underestimate**
- (C) Biased, with a tendency to overestimate

**Solution:** The dotplot reveals that the majority of Jacob's samples contain proportions less than the true value. As such, his sampling method appears to be biased with a tendency to **underestimate**. This could potentially be explained by the fact that college towns (like SLO) tend to have fewer 1bd apartments than non-college towns, as builders know that multi-bedroom apartments are popular among students who can share rooms.

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## Use the following setup for Questions 5 - 9:

Pietro claims to be able to tell the number of blue M&Ms in a pack of M&Ms without ever opening the pack. Among a set of 12 packs of M&Ms, Pietro correctly identified the number of blue M&Ms in 1 of them.

5. How would you describe this study?
- (A) Sample from a Population
  - (B) Observations from a Random Process**

**Solution:** There's no "population" to speak of in this study.

6. What (if any) is the finite population in this study?
- (A) The set of all packs of M&Ms in the world.
  - (B) The set of all blue M&Ms in the world.
  - (C) The set of all people named Pietro in the world.
  - (D) There is no finite population.**

**Solution:** When our data is observations from a random process, we don't have a finite population (i.e. a finite set of observational units of interest). If we like, we can imagine a probability as an *infinite* population proportion, however I'd urge you to think of probabilities more as we defined them, as long-run proportions.

7. What is one observational unit?
- (A) One pack of M&Ms
  - (B) One blue M&M
  - (C) One person
  - (D) One trial in which Pietro tries to identify the number of blue M&Ms in a pack**

**Solution:** When our data is observations from a random process, each observational unit is one trial of the process.

8. What is the sample size?
- (A) 1/12                      (B) 1                      **(C) 12**                      (D) Something Else

**Solution:** When our data is observations from a random process, each observational unit is one trial of the process. The sample size is therefore equal to the number of trials.

9. What is the parameter of interest?
- (A) The proportion of a pack of M&Ms that are blue
  - (B) The proportion of all packs of M&Ms in the world that contain at least one blue M&M
  - (C) The proportion of twelve packs for which Pietro correctly identifies the number of blue M&Ms
  - (D) The long-run proportion of times Pietro correctly guesses the number of blue M&Ms in a pack**

**Solution:** We're essentially interested in the *probability* that Pietro correctly guesses the number of blue M&Ms in a pack without looking at it. The precise definition of a probability is a long-run proportion.

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## Questions 10 - 11 are unrelated:

**10.** The probability that a particular bus arrives at a stop late is 0.3. Which of the following best describes the interpretation of the value of 0.3?

- (A) Out of the next 30 times the bus arrives at the bus stop, it will be late exactly 10 times.
- (B) Out of all the times the bus arrives at the bus stop tomorrow, it will be late exactly 30% of the time.
- (C) Among all the times the bus arrives at the bus stop over the next month, we'd expect it to be late around 30% of the time.**

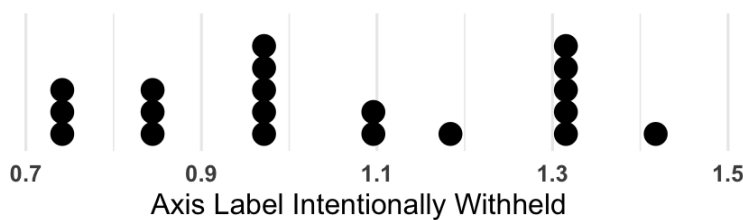
**Solution:** Answer choice (A) is too strong - probabilities are not guarantees! Answer choice (B) seems tempting, but the key is to remember that probabilities are *long-run* proportions; the statement in choice (B) is missing that crucial "long-run" phrasing and is also still too strong ("exactly"). By process of elimination, this leaves choice (C) as the best interpretation.

Now, notice that choice (C) doesn't explicitly say "long-run." This is intentional - I wanted to also show you an equally acceptable interpretation of a probability as giving us a rough sense of what to *expect* in a large population.

To summarize, the main things I'd like you to keep in mind when interpreting probabilities are:

- They are not guarantees (don't use them to make statements that are too strong/definitive)
- They are long-run proportions, which in turn give us a sense of what to expect in a large population

**11.** Marissa takes several samples of 12 cats and records the standard deviation of weights in each sample. She then uses her sample standard deviation values to create the following dotplot:



**Figure 2:** Dotplot of Marissa's Data.

What does one dot on this dotplot represent?

- (A) The weight of one cat
- (B) One cat
- (C) The standard deviation of a sample of 12 cat weights**
- (D) The standard deviation among all cats in Marissa's sampling frame
- (E) Something else